



Mini project report on

Wildlife Conservation Management System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering UE23CS351A

– DBMS Project

Submitted by:

Kashyap K

PES2UG23CS263

Rohan S Nayak

PES2UG24CS820

under the guidance of

Prof. Shilpa S

Assistant Professor

PES University

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India



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CERTIFICATE

This is to certify that the mini project entitled

WildTrack – Wildlife Conservation Management System

is a bonafide work carried out by

Kashyap K

PES2UG23CS263

Rohan S Nayak

PES2UG24CS820

In partial fulfilment for the completion of fifth semester DBMS Project (UE23CS351A) in the Program of Study Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2022 – DEC. 2022. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature

Prof. Shilpa S

Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **WildTrack – Wildlife Conservation Management System** has been carried out by us under the guidance of **Prof. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2023.

Kashyap K
Rohan S Nayak

PES2UG23CS263
PES2UG24CS820

Kashyap
Rohan

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ABSTRACT

The *WildTrack-Wildlife Conservation Management System* is a database-driven platform designed to efficiently manage information about wildlife species, habitats, conservation programs, staff, and sponsorships. The project aims to aid wildlife research and protection efforts through organized data storage, analysis, and visualization.

The system integrates MySQL for backend database management and a Flask + HTML dashboard for front-end visualization. Key features include population tracking, automated updates through triggers, stored procedures for habitat-based queries, and functions for data aggregation such as sponsorship totals and species counts. The project demonstrates the use of SQL concepts including DDL, DML, joins, nested queries, aggregation, stored procedures, and triggers.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	7
2.	PROBLEM DEFINITION	8
3.	ER MODEL	9
4.	ER TO RELATIONAL MAPPING	10
5.	DDL STATEMENTS	12
6.	DML STATEMENTS	16
7.	QUERIES (SIMPLE QUERY AND UPDATE AND DELETE OPERATION, CORRELATED QUERY AND NESTED QUERY)	20
8.	STORED PROCEDURE, FUNCTIONS AND TRIGGERS	24
9.	FRONT END DEVELOPMENT	25
	REFERENCES/BIBLIOGRAPHY	27
	APPENDIX A DEFINITIONS, ACRONYMS AND ABBREVIATIONS	28
	GITHUB LINK	29

1. INTRODUCTION

In recent years, the increasing threats to biodiversity and wildlife have made effective conservation management a global priority. Conservation authorities and researchers often face challenges in maintaining accurate, centralized, and up-to-date data regarding species, habitats, medical records, conservation programs, and sponsorships. Managing such vast and interrelated data manually can lead to inconsistencies, redundancy, and inefficiencies in monitoring and decision-making.

To address these challenges, the *WildTrack – Wildlife Conservation Management System* has been developed as a robust, database-driven solution. The project leverages the capabilities of Relational Database Management Systems (RDBMS) to efficiently store, organize, and manipulate complex conservation-related data. It focuses on building a well-structured schema that represents real-world entities like Species, Habitat, Staff, Conservation Programs, Researchers, Visitors, and Sponsorships, along with their relationships.

The main objective of WildTrack is to provide a centralized digital platform where all data related to wildlife management can be stored, retrieved, and analyzed seamlessly. The system helps users perform key operations such as tracking endangered species, recording medical treatments, analyzing migration records, monitoring conservation budgets, and evaluating research activities.

From a technical standpoint, the project demonstrates the complete database design process, starting from Entity–Relationship (ER) modeling, to relational mapping, table creation using DDL commands, data population using DML statements, and performing queries (simple, nested, and correlated) to extract valuable insights. Furthermore, the inclusion of stored procedures, triggers, and functions automates routine tasks like updating population counts or recalculating species status after rescue operations.

The system also features a modern Flask-based web interface that connects the backend MySQL database to a responsive dashboard built using HTML, CSS. This interface visualizes statistics such as total species, endangered counts, and conservation budgets, allowing real-time insights into conservation activities.

Overall, the WildTrack system embodies the integration of database theory with practical application. It provides an efficient model for wildlife conservation data management while showcasing the implementation of fundamental and advanced DBMS concepts. Through this project, the importance of structured data management, relational integrity, and automation in the field of environmental conservation is effectively demonstrated.

2. PROBLEM DEFINITION

Wildlife conservation involves the continuous monitoring and management of diverse ecosystems, endangered species, conservation programs, and environmental research activities. However, in most conservation initiatives, data is often stored in fragmented formats such as spreadsheets, documents, or isolated systems. This leads to several issues, including data redundancy, inconsistency, difficulty in retrieval, and lack of real-time insights.

Additionally, conservation organizations need to manage multiple entities simultaneously—such as *species health records, migration patterns, habitat details, staff assignments, sponsorships, and rescue operations*. Without a unified system, correlating and analyzing these datasets becomes challenging and time-consuming. As a result, decision-making processes regarding endangered species protection, resource allocation, and program evaluation become inefficient.

The core problem is the absence of a centralized, integrated, and automated data management system that can handle the complex relationships among various aspects of wildlife conservation. Manual record-keeping cannot scale to handle large datasets or ensure data accuracy, especially when dealing with interdependent entities such as species and their habitats or researchers and their studies.

To overcome these challenges, there is a need for a robust Database Management System (DBMS) that can efficiently:

- Store and manage interconnected wildlife data (species, habitats, staff, researchers, sponsors, etc.)
- Maintain data consistency and reduce redundancy through relational modeling
- Support analytical queries and reports for research and decision-making
- Automate updates and calculations through triggers, functions, and stored procedures
- Provide a user-friendly interface for visualization and data interaction

The WildTrack – Wildlife Conservation Management System has been developed to solve these issues by implementing a structured database solution combined with an interactive front-end dashboard. This system provides an end-to-end approach to wildlife data management—starting from creation, maintenance, and querying, to visualization—helping researchers and conservation authorities make data-driven decisions for protecting biodiversity.

3. ER MODEL

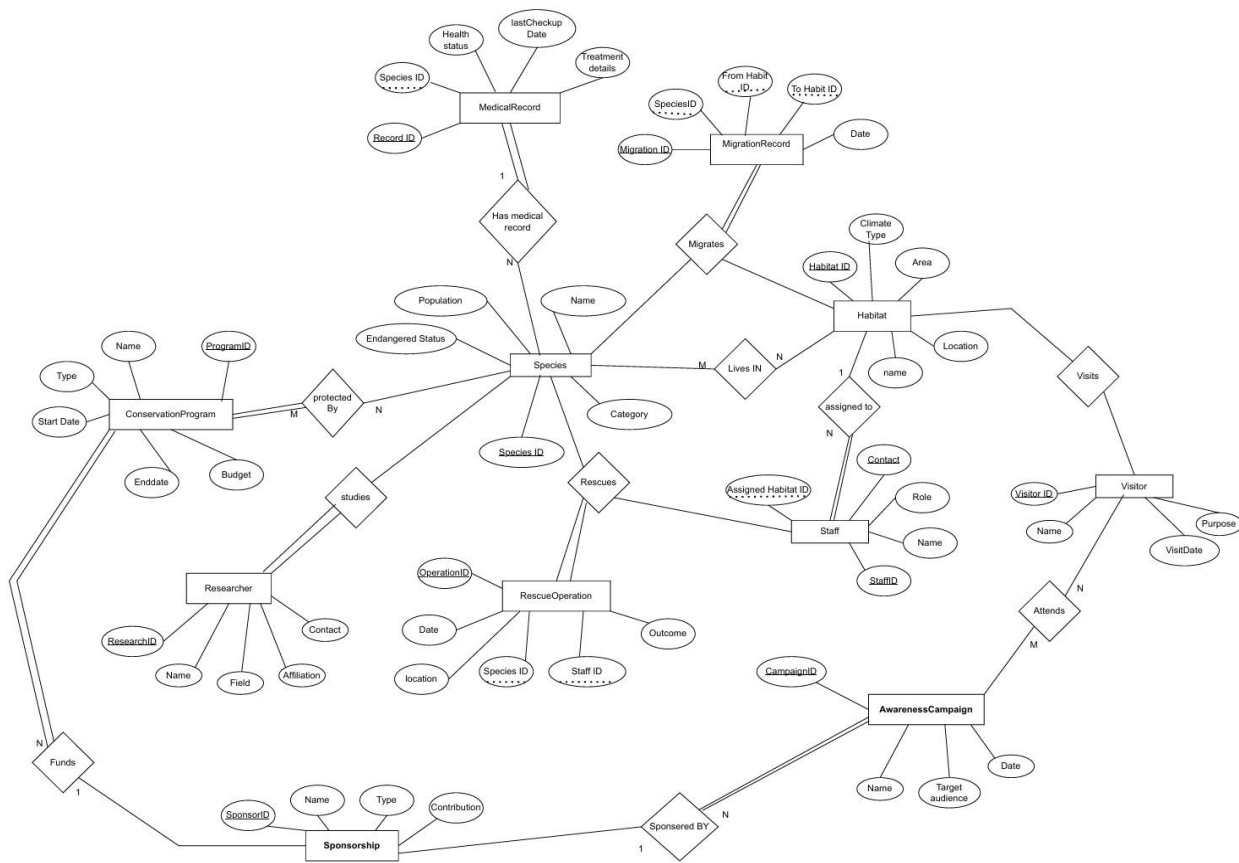
The Entity–Relationship (ER) model for the WildTrack – Wildlife Conservation Management System represents the conceptual structure of the database, capturing the key entities involved in wildlife conservation and the relationships among them.

The main entities identified are:

- Species – Represents different animal species with details such as category, population, and endangered status.
- Habitat – Stores information about various wildlife habitats and their characteristics.
- Staff – Contains details about personnel assigned to different conservation activities.
- ConservationProgram – Tracks ongoing and completed conservation programs and their associated species.
- Researcher – Represents scientists and researchers conducting studies on wildlife.
- Sponsorship – Records details of organizations or individuals funding conservation programs.
- RescueOperation – Stores records of rescue missions conducted by staff for endangered species.
- MedicalRecord – Maintains medical and health details of animals under observation.
- MigrationRecord – Tracks migration movements of species between habitats.
- Visitor – Contains data about individuals visiting conservation sites.
- AwarenessCampaign – Stores details about campaigns aimed at wildlife awareness and education.

The key relationships between these entities are:

- A Species lives in one or more Habitats (LivesIn).
- A Species is protected by multiple Staff members (ProtectedBy).
- A Researcher studies one or more Species (Studies).
- A Species is part of a ConservationProgram (SpeciesID as FK).
- A Visitor attends one or more AwarenessCampaigns (Attends).
- A ConservationProgram is funded by a Sponsorship (Funds).
- A Species has multiple MedicalRecords and MigrationRecords.
- A Staff member performs RescueOperations for Species.



4. ER TO RELATIONAL MAPPING

The ER model has been transformed into a relational database schema using MySQL. Each entity has been mapped to a corresponding table, with primary and foreign keys ensuring referential integrity.

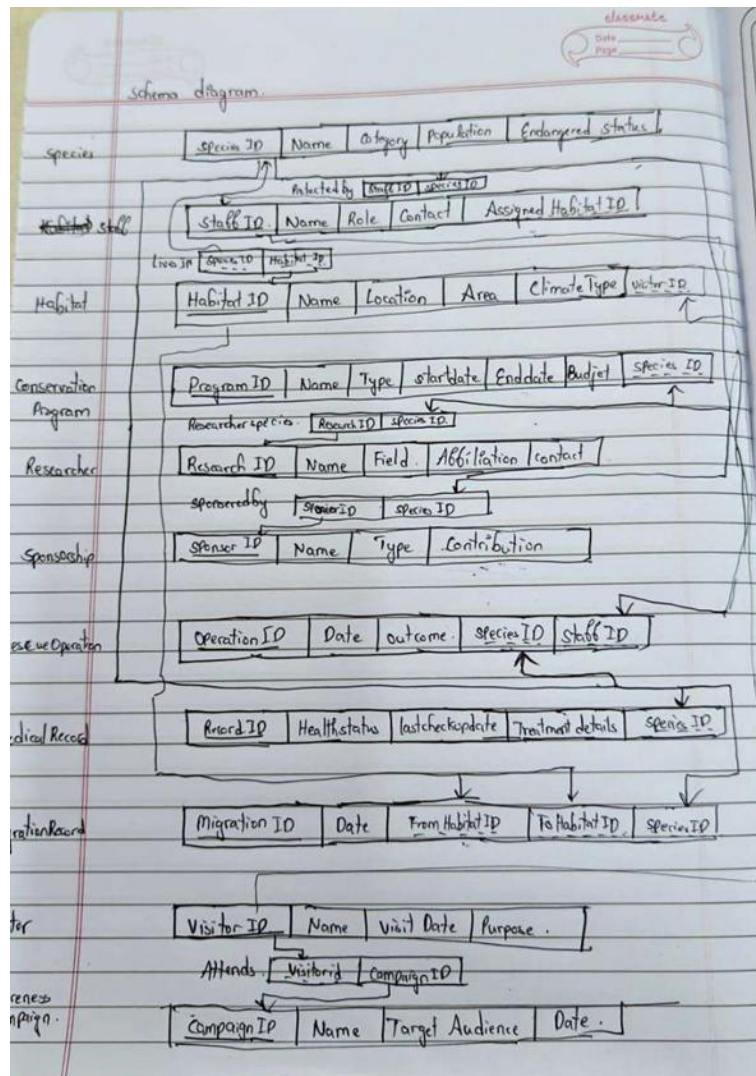
Mapped Tables:

- Species(SpeciesID, Name, Category, EndangeredStatus, Population)
- Habitat(HabitatID, Name, Location, Area, Climate, Type)
- Staff(StaffID, Name, Role, Contact, AssignedHabitatID [FK → Habitat.HabitatID]*)
- ConservationProgram(ProgramID, Name, Type, Budget, StartDate, EndDate, SpeciesID [FK → Species.SpeciesID]*)
- Researcher(ResearchID, Name, Field, Affiliation, Contact)
- Sponsorship(SponsorID, Name, Type, Contribution, ProgramID [FK → ConservationProgram.ProgramID]*)
- MedicalRecord(RecordID, SpeciesID [FK], HealthStatus, LastCheckupDate, TreatmentDetails)
- Visitor(VisitorID, Name, VisitDate, Purpose)
- MigrationRecord(MigrationID, SpeciesID [FK], FromHabitatID [FK], ToHabitatID [FK], Date)
- AwarenessCampaign(CampaignID, Name, TargetAudience, Date)
- LivesIn(SpeciesID [FK], HabitatID [FK])
- ProtectedBy(SpeciesID [FK], StaffID [FK])
- Studies(ResearchID [FK], SpeciesID [FK])

- Attends(VisitorID [FK], CampaignID [FK])
- RescueOperation(OperationID, SpeciesID [FK], StaffID [FK], Date, Outcome)

All many-to-many relationships have been implemented using junction tables such as *LivesIn*, *ProtectedBy*, *Studies*, and *Attends*. Each junction table includes foreign keys referring to both participating entities, ensuring relational consistency.

4.1 COMPLETE DIAGRAM OF RELATIONAL MAPPING



5. DDL STATEMENTS

Query: create database

Wildlife_Conservation;

Use Wildlife_Conservation;

CREATE TABLE Species (

SpeciesID INT PRIMARY KEY,

Name VARCHAR(100) NOT NULL,

Category VARCHAR(50),

EndangeredStatus VARCHAR(50),

Population INT

);

CREATE TABLE Habitat (

HabitatID INT PRIMARY KEY,

Name VARCHAR(100) NOT NULL,

Location VARCHAR(100),

Area DECIMAL(10,2),

Climate VARCHAR(50),

Type VARCHAR(50)

);

CREATE TABLE Staff (

StaffID INT PRIMARY KEY,

Name VARCHAR(100) NOT NULL,

Role VARCHAR(50),

Contact VARCHAR(50),

AssignedHabitatID INT,

FOREIGN KEY (AssignedHabitatID) REFERENCES Habitat(HabitatID)

);

CREATE TABLE ConservationProgram (

ProgramID INT PRIMARY KEY,

Name VARCHAR(100) NOT NULL,

Type VARCHAR(50),

Budget DECIMAL(12,2),

StartDate DATE,

```

    EndDate DATE,
    SpeciesID INT,
    FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID)
);

CREATE TABLE Researcher (
    ResearchID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Field VARCHAR(100),
    Affiliation VARCHAR(100),
    Contact VARCHAR(50)
);

CREATE TABLE Sponsorship (
    SponsorID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Type VARCHAR(50),
    Contribution DECIMAL(12,2),
    ProgramID INT,
    FOREIGN KEY (ProgramID) REFERENCES ConservationProgram(ProgramID)
);

CREATE TABLE MedicalRecord (
    RecordID INT PRIMARY KEY,
    SpeciesID INT,
    HealthStatus VARCHAR(100),
    LastCheckupDate DATE,
    TreatmentDetails TEXT,
    FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID)
);

CREATE TABLE Visitor (
    VisitorID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    VisitDate DATE,
    Purpose VARCHAR(100)
);

CREATE TABLE MigrationRecord (

```

```

MigrationID INT PRIMARY KEY,
SpeciesID INT,
Date DATE,
FromHabitatID INT,
ToHabitatID INT,
FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID),
FOREIGN KEY (FromHabitatID) REFERENCES Habitat(HabitatID),
FOREIGN KEY (ToHabitatID) REFERENCES Habitat(HabitatID)
);

CREATE TABLE AwarenessCampaign (
    CampaignID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    TargetAudience VARCHAR(100),
    Date DATE
);

CREATE TABLE LivesIn (
    SpeciesID INT,
    HabitatID INT,
    FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID),
    FOREIGN KEY (HabitatID) REFERENCES Habitat(HabitatID)
);

CREATE TABLE ProtectedBy (
    SpeciesID INT,
    StaffID INT,
    PRIMARY KEY (SpeciesID, StaffID),
    FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID),
    FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)
);

CREATE TABLE Studies (
    ResearchID INT,
    SpeciesID INT,
    PRIMARY KEY (ResearchID, SpeciesID),
    FOREIGN KEY (ResearchID) REFERENCES Researcher(ResearchID),
    FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID)
);

```

);

CREATE TABLE Attends (

VisitorID INT,

CampaignID INT,

PRIMARY KEY (VisitorID, CampaignID),

FOREIGN KEY (VisitorID) REFERENCES Visitor(VisitorID),

FOREIGN KEY (CampaignID) REFERENCES AwarenessCampaign(CampaignID)

);

CREATE TABLE RescueOperation (

OperationID INT PRIMARY KEY,

SpeciesID INT,

StaffID INT,

Date DATE,

Outcome VARCHAR(100),

FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID),

FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)

);

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	00:08:09	create database Wildlife_Conservation	1 row(s) affected	0.000 sec
2	00:08:09	Use Wildlife_Conservation	0 row(s) affected	0.000 sec
3	00:08:09	CREATE TABLE Species (SpeciesID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Categor...	0 row(s) affected	0.016 sec
4	00:08:09	CREATE TABLE LIVES (SpeciesID INT PRIMARY KEY, HabitatID INT, FOREIGN KEY (SpeciesID) REFERENCES Species(SpeciesID),	0 row(s) affected	0.016 sec
12	00:08:09	CREATE TABLE AwarenessCampaign (CampaignID INT PRIMARY KEY, Name VARCHAR(100) NOT NU...	0 row(s) affected	0.015 sec
13	00:08:09	CREATE TABLE LivesIn (SpeciesID INT, HabitatID INT, FOREIGN KEY (SpeciesID) REFERENCES S...	0 row(s) affected	0.016 sec
14	00:08:09	CREATE TABLE ProtectedBy (SpeciesID INT, StaffID INT, PRIMARY KEY (SpeciesID, StaffID), FO...	0 row(s) affected	0.031 sec
15	00:08:09	CREATE TABLE Studies (ResearchID INT, SpeciesID INT, PRIMARY KEY (ResearchID, SpeciesID), ...	0 row(s) affected	0.016 sec
16	00:08:09	CREATE TABLE Attends (VisitorID INT, CampaignID INT, PRIMARY KEY (VisitorID, CampaignID), F...	0 row(s) affected	0.015 sec
17	00:08:09	CREATE TABLE RescueOperation (OperationID INT PRIMARY KEY, SpeciesID INT, StaffID INT, D...	0 row(s) affected	0.031 sec
9	00:08:09	CREATE TABLE MedicalRecord (RecordID INT PRIMARY KEY, SpeciesID INT, HealthStatus VARCH...	0 row(s) affected	0.016 sec
10	00:08:09	CREATE TABLE Visitor (VisitorID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, VisitDate DA...	0 row(s) affected	0.015 sec
11	00:08:09	CREATE TABLE MigrationRecord (MigrationID INT PRIMARY KEY, SpeciesID INT, Date DATE, Fro...	0 row(s) affected	0.016 sec

6. DML STATEMENTS

Query :

use wildlife_conservation;

INSERT INTO Species (SpeciesID, Name, Category, EndangeredStatus, Population) VALUES

(1,'Bengal Tiger','Mammal','Endangered',120),
(2,'Asian Elephant','Mammal','Vulnerable',350),
(3,'Snow Leopard','Mammal','Endangered',80),
(4,'Indian Rhino','Mammal','Vulnerable',200),
(5,'Olive Ridley Turtle','Reptile','Vulnerable',500),
(6,'Ganges Dolphin','Mammal','Endangered',90),
(7,'Indian Peafowl','Bird','Least Concern',900),
(8,'Red Panda','Mammal','Endangered',60),
(9,'King Cobra','Reptile','Vulnerable',400),
(10,'Lion-tailed Macaque','Mammal','Endangered',120);

INSERT INTO Habitat (HabitatID, Name, Location, Area, Climate, Type) VALUES

(1,'Sundarbans Reserve','West Bengal',9600.50,'Tropical','Forest'),
(2,'Kaziranga Park','Assam',4300.75,'Humid','Grassland'),
(3,'Himalayan Reserve','Ladakh',12000.20,'Cold','Mountain'),
(4,'Chilika Lake','Odisha',1500.00,'Moderate','Wetland'),
(5,'Gir Forest','Gujarat',2200.00,'Dry','Forest'),
(6,'Corbett Park','Uttarakhand',1318.54,'Temperate','Forest'),
(7,'Periyar Sanctuary','Kerala',925.65,'Humid','Rainforest'),
(8,'Ranthambore','Rajasthan',1334.00,'Dry','Forest'),
(9,'Nagarhole','Karnataka',643.39,'Moderate','Forest'),
(10,'Keoladeo Park','Rajasthan',500.00,'Dry','Wetland');

INSERT INTO Staff (StaffID, Name, Role, Contact, AssignedHabitatID) VALUES

(1,'Arun Sharma','Ranger','9876543210',1),
(2,'Meena Das','Veterinarian','9123456789',2),
(3,'Ravi Kumar','Ecologist','9988776655',3),
(4,'Sneha Iyer','Guide','9876501234',4),
(5,'Ramesh Patil','Warden','9876000012',5),
(6,'Vivek Shetty','Forest Officer','9123012301',6),
(7,'Divya Rao','Zoologist','9234567890',7),
(8,'Kiran Singh','Tracker','9345678901',8),
(9,'Pooja Jain','Rescue Worker','9456789012',9),
(10,'Manish Thakur','Research Assistant','9567890123',10);

INSERT INTO ConservationProgram (ProgramID, Name, Type, Budget, StartDate, EndDate, SpeciesID) VALUES

(1,'Project Tiger','Protection',1200000,'2024-01-01','2026-12-31',1),
(2,'Save the Elephants','Research',900000,'2024-03-15','2026-03-14',2),
(3,'Snow Leopard Initiative','Tracking',800000,'2025-01-01','2026-12-31',3),
(4,'Rhino Care Mission','Medical',400000,'2025-04-01','2025-12-31',4),
(5,'Turtle Nest Watch','Awareness',350000,'2025-02-01','2025-12-31',5),
(6,'Ganga Dolphin Project','Research',500000,'2025-03-01','2026-02-28',6),

(7,'Peafowl Protection','Habitat Restoration',250000,'2024-08-01','2025-07-31',7),
(8,'Red Panda Revival','Breeding',600000,'2025-05-01','2026-05-01',8),
(9,'King Cobra Safety','Awareness',200000,'2025-06-01','2025-12-01',9),
(10,'Macaque Monitoring','Tracking',300000,'2024-09-01','2026-08-31',10);

INSERT INTO Researcher (ResearchID, Name, Field, Affiliation, Contact) VALUES

(1,'Dr. Kavya Menon','Zoology','PES University','9999998888'),
(2,'Dr. Arjun Rao','Ecology','IISc Bangalore','8888887777'),
(3,'Dr. Neha Patel','Marine Biology','NCBS','7777776666'),
(4,'Dr. Ankit Jain','Ornithology','BNHS','6666665555'),
(5,'Dr. Meera Shah','Wildlife Medicine','WII Dehradun','9999900000'),
(6,'Dr. Keshav Reddy','Mammalogy','IIT Bombay','9898989898'),
(7,'Dr. Leela Krishnan','Herpetology','ZSI India','9797979797'),
(8,'Dr. Ananya Roy','Conservation Biology','WWF India','9696969696'),
(9,'Dr. Rajesh Naik','Veterinary Science','IISER Pune','9595959595'),
(10,'Dr. Tanuja Bhat','Environmental Science','NIT Trichy','9494949494');

INSERT INTO Sponsorship (SponsorID, Name, Type, Contribution, ProgramID) VALUES

(1,'WWF India','NGO',500000,1),
(2,'Tata Trusts','Corporate',300000,2),
(3,'UNDP','International',200000,3),
(4,'Forest Dept. India','Government',400000,4),
(5,'Reliance Foundation','Corporate',250000,5),
(6,'Amazon Cares','Corporate',220000,6),
(7,'Google India','Corporate',310000,7),
(8,'IUCN','International',450000,8),
(9,'Save Earth Org','NGO',180000,9),
(10,'Infosys Foundation','Corporate',270000,10);

INSERT INTO MedicalRecord (RecordID, SpeciesID, HealthStatus, LastCheckupDate, TreatmentDetails) VALUES

(1,1,'Healthy','2025-08-10','Routine vaccination done'),
(2,2,'Injured','2025-09-01','Leg wound treatment ongoing'),
(3,3,'Stable','2025-07-25','Monitoring oxygen intake'),
(4,4,'Recovered','2025-06-20','Post-poaching recovery'),
(5,5,'Healthy','2025-08-15','Nesting inspection done'),
(6,6,'Critical','2025-09-10','Respiratory infection treatment'),
(7,7,'Healthy','2025-07-19','Regular check'),
(8,8,'Weak','2025-10-01','Low appetite; supplements given'),
(9,9,'Recovered','2025-08-29','Antivenom tests completed'),
(10,10,'Healthy','2025-09-18','Routine health check');

INSERT INTO Visitor (VisitorID, Name, VisitDate, Purpose) VALUES

(1,'Rohit Verma','2025-08-01','Education'),
(2,'Priya Nair','2025-08-10','Tourism'),
(3,'Ananya Singh','2025-09-05','Research'),
(4,'Amit Joshi','2025-09-15','Photography'),
(5,'Kiran Rao','2025-10-01','Awareness'),
(6,'Suresh Mehta','2025-07-20','Volunteer');

(7,'Nidhi Patel','2025-09-22','Campaign'),
(8,'Aditya Rao','2025-10-05','Wildlife Workshop'),
(9,'Riya Sen','2025-09-30','Education'),
(10,'Manoj Gupta','2025-08-25','Tourism');

INSERT INTO MigrationRecord (MigrationID, SpeciesID, Date, FromHabitatID, ToHabitatID) VALUES
(1,3,'2025-07-01',3,2),
(2,1,'2025-05-20',1,8),
(3,2,'2025-08-15',2,1),
(4,4,'2025-03-10',5,3),
(5,5,'2025-09-22',4,9),
(6,6,'2025-08-05',6,1),
(7,7,'2025-07-10',10,7),
(8,8,'2025-08-18',7,6),
(9,9,'2025-10-10',9,5),
(10,10,'2025-06-25',8,2);

INSERT INTO LivesIn (SpeciesID, HabitatID) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),
(6,6),(7,7),(8,8),(9,9),(10,10);

INSERT INTO ProtectedBy (SpeciesID, StaffID) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),
(6,6),(7,7),(8,8),(9,9),(10,10);

INSERT INTO Studies (ResearchID, SpeciesID) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),
(6,6),(7,7),(8,8),(9,9),(10,10);

INSERT INTO AwarenessCampaign (CampaignID, Name, TargetAudience, Date) VALUES
(1,'Save Tigers','General Public','2025-09-10'),
(2,'Protect Elephants','School Students','2025-08-15'),
(3,'Snow Awareness','Local Communities','2025-07-05'),
(4,'Rhino Rally','Wildlife Enthusiasts','2025-09-20'),
(5,'Turtle Week','Tourists','2025-10-05'),
(6,'Dolphin Day','Fishermen','2025-06-08'),
(7,'Birds & Biodiversity','Researchers','2025-09-12'),
(8,'Panda Parade','Children','2025-07-30'),
(9,'Snake Safety','Villagers','2025-10-12'),
(10,'Monkey Mind','Forest Rangers','2025-09-25');

INSERT INTO Attends (VisitorID, CampaignID) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),
(6,6),(7,7),(8,8),(9,9),(10,10);

INSERT INTO RescueOperation (OperationID, SpeciesID, StaffID, Date, Outcome) VALUES
(1, 1, 1, '2025-07-12', 'Tiger relocated safely'),
(2, 2, 2, '2025-08-04', 'Elephant freed from trap'),

(3, 3, 3, '2025-09-01', 'Snow leopard tagged and released'),
(4, 4, 4, '2025-07-19', 'Rhino treated for wound'),
(5, 5, 5, '2025-08-23', 'Turtle saved from fishing net'),
(6, 6, 6, '2025-09-02', 'Dolphin guided back to river'),
(7, 7, 7, '2025-08-28', 'Peafowl rescued from road accident'),
(8, 8, 8, '2025-10-03', 'Red panda safely moved'),
(9, 9, 9, '2025-09-15', 'King cobra rescued from village'),
(10, 10, 10, '2025-08-10', 'Macaque released after medical check');

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
2	00:11:26	INSERT INTO Species (SpeciesID, Name, Category, EndangeredStatus, Population) VALUES (1,'Bengal Tiger',...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
3	00:11:26	INSERT INTO Habitat (HabitatID, Name, Location, Area, Climate, Type) VALUES (1,'Sundarbans Reserve','We...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
4	00:11:26	INSERT INTO Staff (StaffID, Name, Role, Contact, AssignedHabitatID) VALUES (1,'Arun Shama','Ranger','987...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
5	00:11:26	INSERT INTO ConservationProgram (ProgramID, Name, Type, Budget, StartDate, EndDate, SpeciesID) VALUE...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec
6	00:11:26	INSERT INTO Researcher (ResearchID, Name, Field, Affiliation, Contact) VALUES (1,'Dr. Kavya Menon','Zoolo...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
7	00:11:26	INSERT INTO Sponsorship (SponsorID, Name, Type, Contribution, ProgramID) VALUES (1,'WWF India','NGO'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
8	00:11:26	INSERT INTO MedicalRecord (RecordID, SpeciesID, HealthStatus, LastCheckupDate, TreatmentDetails) VAL...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
9	00:11:26	INSERT INTO Visitor (VisitorID, Name, VisitDate, Purpose) VALUES (1,'Rohit Vema','2025-08-01','Education'), (...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
10	00:11:26	INSERT INTO MigrationRecord (MigrationID, SpeciesID, Date, FromHabitatID, ToHabitatID) VALUES (1,3,202...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
11	00:11:26	INSERT INTO LivesIn (SpeciesID, HabitatID) VALUES (1,1),(2,2),(3,3),(4,4),(5,5), (6,6),(7,7),(8,8),(9,9),(10,10)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
12	00:11:26	INSERT INTO ProtectedBy (SpeciesID, StaffID) VALUES (1,1),(2,2),(3,3),(4,4),(5,5), (6,6),(7,7),(8,8),(9,9),(10,10)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
13	00:11:26	INSERT INTO Studies (ResearchID, SpeciesID) VALUES (1,1),(2,2),(3,3),(4,4),(5,5), (6,6),(7,7),(8,8),(9,9),(10,10)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
14	00:11:26	INSERT INTO AwarenessCampaign (CampaignID, Name, TargetAudience, Date) VALUES (1,'Save Tigers','Ge...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
15	00:11:26	INSERT INTO NewsArticle (ArticleID, SpeciesID) VALUES (1,1),(2,2),(3,3),(4,4),(5,5), (6,6),(7,7),(8,8),(9,9),(10,10)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

152

153 • select * from species;

154

Result Grid				
Filter Rows:				
Edit: Export/Import: Wrap Cell Content:				
SpeciesID	Name	Category	EndangeredStatus	Population
1	Bengal Tiger	Mammal	Endangered	120
2	Asian Elephant	Mammal	Vulnerable	350
3	Snow Leopard	Mammal	Endangered	80
4	Indian Rhino	Mammal	Vulnerable	200
5	Olive Ridley Turtle	Reptile	Vulnerable	500
6	Ganges Dolphin	Mammal	Endangered	90
7	Indian Peafowl	Bird	Least Concern	900
8	Red Panda	Mammal	Endangered	60
9	King Cobra	Reptile	Vulnerable	400
10	Lion-tailed Macaque	Mammal	Endangered	120
NULL	NULL	NULL	NULL	NULL

153 • select * from RescueOperation;

Result Grid				
Filter Rows:				
Edit: Export/Import: Wrap Cell Content:				
OperationID	SpeciesID	StaffID	Date	Outcome
1	1	1	2025-07-12	Tiger relocated safely
2	2	2	2025-08-04	Elephant freed from trap
3	3	3	2025-09-01	Snow leopard tagged and released
4	4	4	2025-07-19	Rhino treated for wound
5	5	5	2025-08-23	Turtle saved from fishing net
6	6	6	2025-09-02	Dolphin guided back to river
7	7	7	2025-08-28	Peafowl rescued from road accident
8	8	8	2025-10-03	Red panda safely moved
9	9	9	2025-09-15	King cobra rescued from village
10	10	10	2025-08-10	Macaque released after medical check
NULL	NULL	NULL	NULL	NULL

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGGREGATE

-- Aggregate queries

-- avg budget of all programs

```
SELECT AVG(Budget) AS AverageBudget  
FROM ConservationProgram;
```



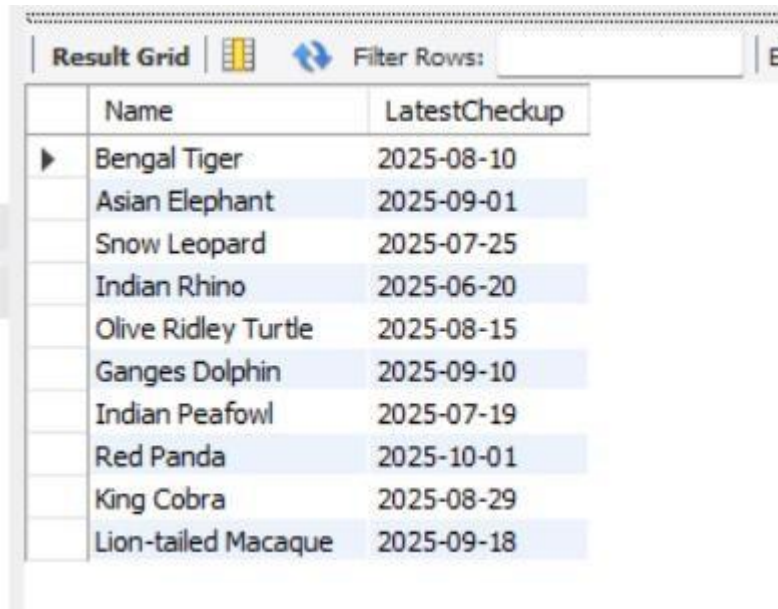
The screenshot shows a database query result grid. The toolbar includes 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The query result is a single row with the column 'AverageBudget' and the value '550000.000000'.

AverageBudget
550000.000000

-- Group By

-- most recently checked up species

```
SELECT s.Name, MAX(m.LastCheckupDate) AS LatestCheckup  
FROM Species s  
JOIN MedicalRecord m ON s.SpeciesID = m.SpeciesID  
GROUP BY s.SpeciesID, s.Name;
```

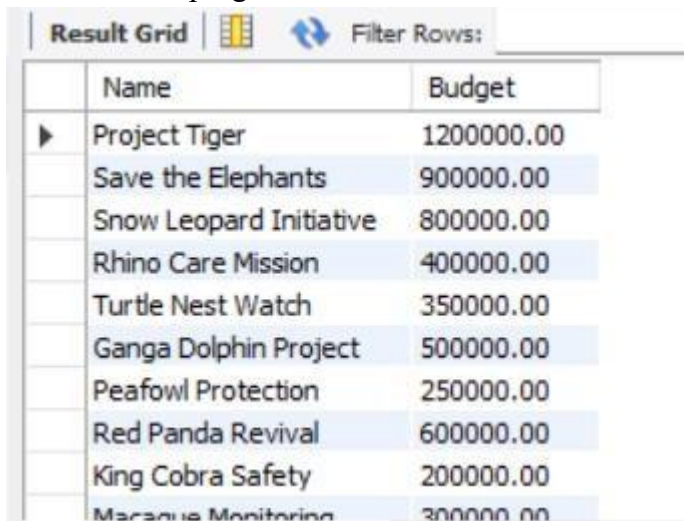


The screenshot shows a database query result grid. The toolbar includes 'Result Grid', 'Filter Rows', and 'Export'. The query result is a table with two columns: 'Name' and 'LatestCheckup'. The data is as follows:

Name	LatestCheckup
Bengal Tiger	2025-08-10
Asian Elephant	2025-09-01
Snow Leopard	2025-07-25
Indian Rhino	2025-06-20
Olive Ridley Turtle	2025-08-15
Ganges Dolphin	2025-09-10
Indian Peafowl	2025-07-19
Red Panda	2025-10-01
King Cobra	2025-08-29
Lion-tailed Macaque	2025-09-18

7.2 UPDATE OPERATION

Conservation program table:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'Name' and 'Budget'. There are 11 rows of data. The first row, 'Project Tiger', has a budget of 1200000.00. The other rows have budgets ranging from 200000.00 to 900000.00. The interface includes a 'Filter Rows' button and a 'Filter Rows:' text box.

	Name	Budget
▶	Project Tiger	1200000.00
	Save the Elephants	900000.00
	Snow Leopard Initiative	800000.00
	Rhino Care Mission	400000.00
	Turtle Nest Watch	350000.00
	Ganga Dolphin Project	500000.00
	Peafowl Protection	250000.00
	Red Panda Revival	600000.00
	King Cobra Safety	200000.00
	Macaque Monitoring	300000.00

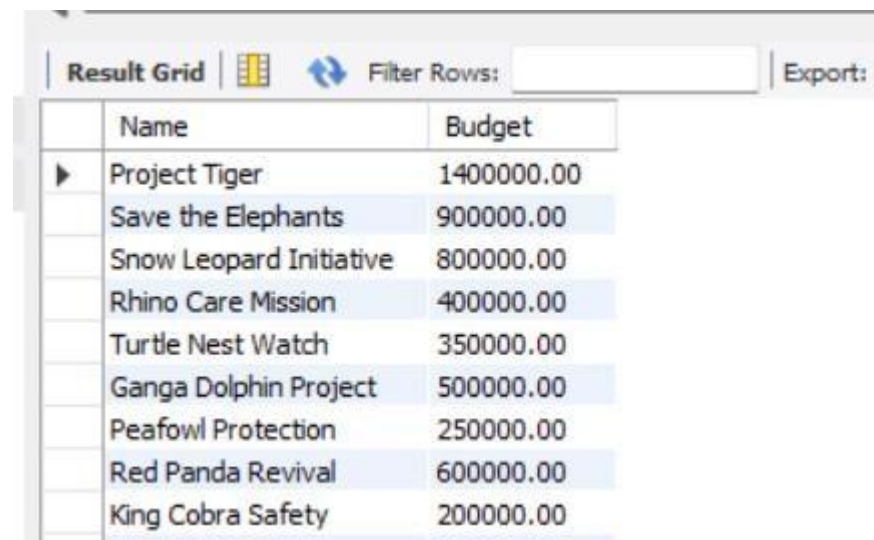
Query:

```
UPDATE ConservationProgram
```

```
SET Budget = Budget + 200000
```

```
WHERE Name = 'Project Tiger';
```

Result table:



The screenshot shows the same database interface as before, but the 'Project Tiger' row now has a budget of 1400000.00, reflecting the update. The other rows remain unchanged. The interface includes a 'Filter Rows' button, a 'Filter Rows:' text box, and an 'Export:' button.

	Name	Budget
▶	Project Tiger	1400000.00
	Save the Elephants	900000.00
	Snow Leopard Initiative	800000.00
	Rhino Care Mission	400000.00
	Turtle Nest Watch	350000.00
	Ganga Dolphin Project	500000.00
	Peafowl Protection	250000.00
	Red Panda Revival	600000.00
	King Cobra Safety	200000.00

7.3 DELETE OPERATION

Visitor table:

RecordID	SpeciesID	HealthStatus	LastCheckupDate	TreatmentDetails
1	1	Healthy	2025-08-10	Routine vaccination done
2	2	Injured	2025-09-01	Leg wound treatment ongoing
3	3	Stable	2025-07-25	Monitoring oxygen intake
4	4	Recovered	2025-06-20	Post-poaching recovery
5	5	Healthy	2025-08-15	Nesting inspection done
6	6	Critical	2025-09-10	Respiratory infection treatment
7	7	Healthy	2025-07-19	Regular check
8	8	Weak	2025-10-01	Low appetite; supplements given
9	9	Recovered	2025-08-29	Antivenom tests completed
10	10	Healthy	2025-09-18	Routine health check
NULL	NULL	NULL	NULL	NULL

Query:

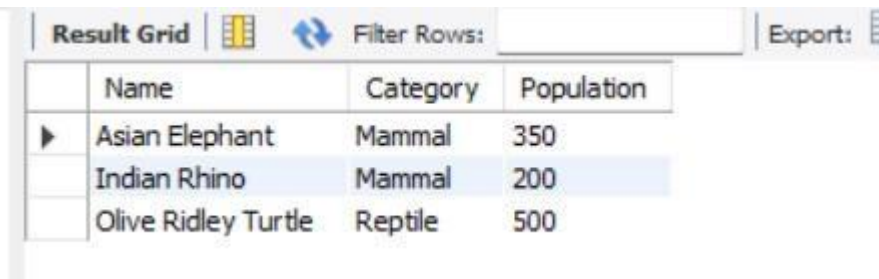
```
DELETE FROM MedicalRecord
WHERE LastCheckupDate < '2025-07-01';
```

Resulting Table: (4th Entry is deleted)

RecordID	SpeciesID	HealthStatus	LastCheckupDate	TreatmentDetails
1	1	Healthy	2025-08-10	Routine vaccination done
2	2	Injured	2025-09-01	Leg wound treatment ongoing
3	3	Stable	2025-07-25	Monitoring oxygen intake
5	5	Healthy	2025-08-15	Nesting inspection done
6	6	Critical	2025-09-10	Respiratory infection treatment
7	7	Healthy	2025-07-19	Regular check
8	8	Weak	2025-10-01	Low appetite; supplements given
9	9	Recovered	2025-08-29	Antivenom tests completed
10	10	Healthy	2025-09-18	Routine health check
NULL	NULL	NULL	NULL	NULL

7.4 CORRELATED QUERY

```
SELECT Name, Category, Population
FROM Species s
WHERE Population > (
    SELECT AVG(Population)
    FROM Species
    WHERE Category = s.Category
);
```



	Name	Category	Population
▶	Asian Elephant	Mammal	350
	Indian Rhino	Mammal	200
	Olive Ridley Turtle	Reptile	500

7.5 NESTED QUERY

```
SELECT Name
FROM Species
WHERE SpeciesID IN (
    SELECT SpeciesID
    FROM ConservationProgram
    WHERE Budget > 1000000
);
```



	Name
▶	Bengal Tiger

8. STORED PROCEDURES, FUCNTIONS AND TRIGGERS

8.1 STORED PROCEDURES OR FUNCTIONS

Adding ConservationProgram (Procedure)

```
DELIMITER //
CREATE PROCEDURE AddConservationProgram (
    IN p_Name VARCHAR(100),
    IN p_Type VARCHAR(50),
    IN p_Budget DECIMAL(12,2),
    IN p_Start DATE,
    IN p_End DATE,
    IN p_SpeciesID INT
)
BEGIN
    INSERT INTO ConservationProgram (Name, Type, Budget, StartDate, EndDate, SpeciesID)
VALUES (p_Name, p_Type, p_Budget, p_Start, p_End, p_SpeciesID); END //
DELIMITER ;
```

Function: Calculating Species By category

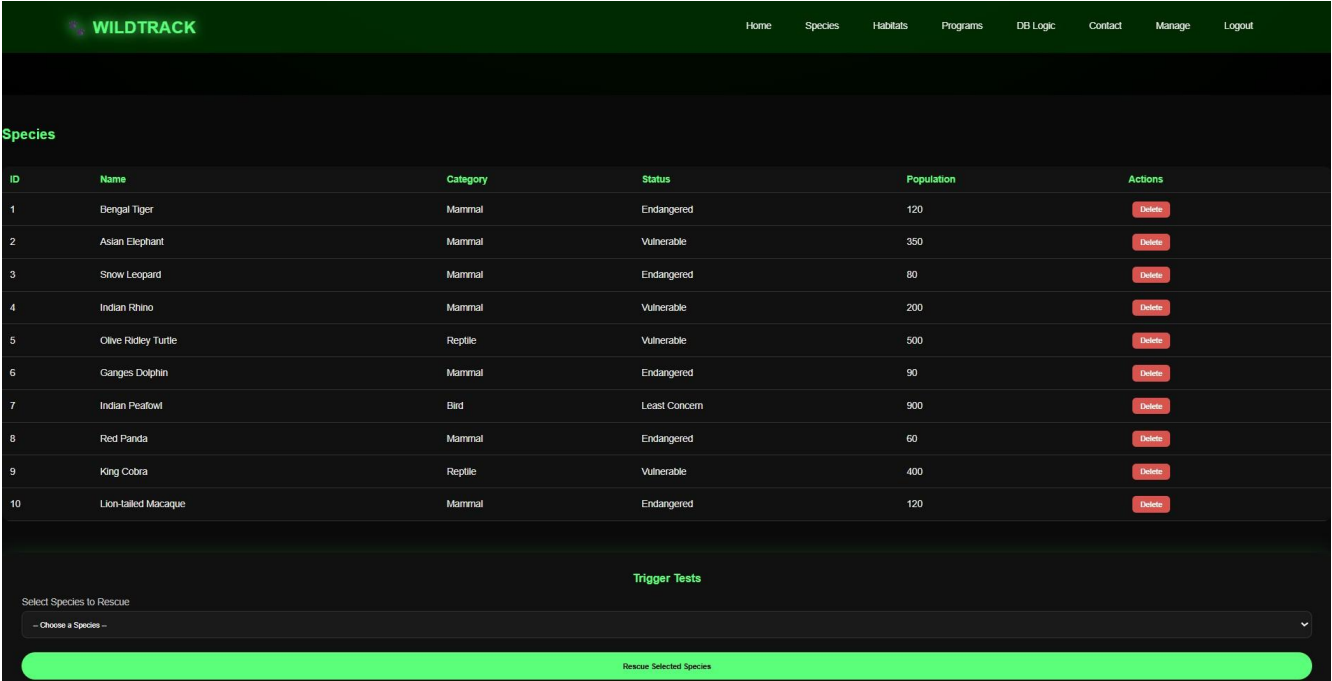
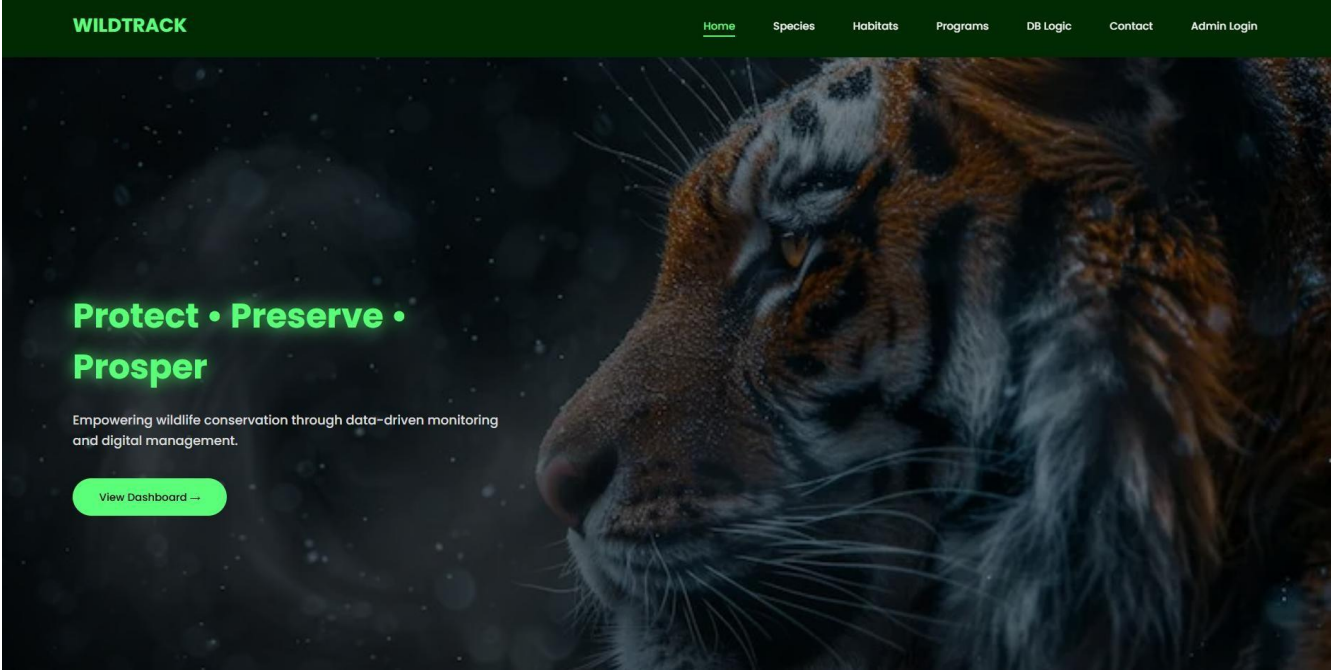
```
DELIMITER //
CREATE FUNCTION SpeciesCountByCategory(cat VARCHAR(50))
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE cnt INT;
    SELECT COUNT(*) INTO cnt FROM Species WHERE Category = cat;
    RETURN cnt;
END //
DELIMITER ;
```

8.2 TRIGGERS

Updating Population after rescue:

```
DELIMITER //
CREATE TRIGGER UpdatePopulationAfterRescue
AFTER INSERT ON RescueOperation
FOR EACH ROW
BEGIN
    UPDATE Species
SET Population = Population + 1
WHERE SpeciesID = NEW.SpeciesID;
END;
//
```


9. FRONT END DEVELOPEMNT



WILDTRACK

Home

Species

Habitats

Programs

DB Logic

Contact

Manage

Logout

Conservation Program Management

Add New Program

Program Name

Program Type

Budget (₹)

dd-mm-yyyy

dd-mm-yyyy

Select Species

Add Program

ID	Name	Type	Budget (₹)	Start	End	Species	Action
10	Macaque Monitoring	Tracking	300000.00	2024-09-01	2026-08-31	Lion-tailed Macaque	Delete
9	King Cobra Safety	Awareness	200000.00	2025-06-01	2025-12-01	King Cobra	Delete
8	Red Panda Revival	Breeding	600000.00	2025-05-01	2026-05-01	Red Panda	Delete
7	Peafowl Protection	Habitat Restoration	250000.00	2024-08-01	2025-07-31	Indian Peafowl	Delete
6	Ganga Dolphin Project	Research	500000.00	2025-03-01	2026-02-28	Ganges Dolphin	Delete
5	Turtle Nest Watch	Awareness	350000.00	2025-02-01	2025-12-31	Olive Ridley Turtle	Delete
4	Rhino Care Mission	Medical	400000.00	2025-04-01	2025-12-31	Indian Rhino	Delete
3	Snow Leopard Initiative	Tracking	800000.00	2025-01-01	2026-12-31	Snow Leopard	Delete
2	Save the Elephants	Research	900000.00	2024-03-15	2026-03-14	Asian Elephant	Delete

WILDTRACK

Home

Species

Habitats

Programs

DB Logic

Contact

Manage

Logout

Total Species

10

Endangered Species

5

Total Sponsorship

₹3080000.00

Conservation Programs

10

Recent Conservation Efforts

Program	Species	Budget	Start Date
King Cobra Safety	King Cobra	₹200000.00	2025-06-01
Red Panda Revival	Red Panda	₹600000.00	2025-05-01
Rhino Care Mission	Indian Rhino	₹400000.00	2025-04-01
Ganga Dolphin Project	Ganges Dolphin	₹500000.00	2025-03-01
Turtle Nest Watch	Olive Ridley Turtle	₹350000.00	2025-02-01

Developed By Kashyap K And Rohan S Nayak

PES University | UE23CS351A - DBMS

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<https://dev.mysql.com/doc/>
2. **W3Schools.** *MySQL Tutorial.*
<https://www.w3schools.com/mysql/>
3. **Flask Documentation.** *Flask Web Framework for Python.*
<https://flask.palletsprojects.com/>
4. **Mozilla Developer Network (MDN).** *HTML, CSS, and JavaScript References.*
<https://developer.mozilla.org/>

APPENDIX – DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term / Acronym	Definition / Description
SQL	Structured Query Language – used to create and manage relational databases.
DDL	Data Definition Language – SQL commands for creating and altering database structure (e.g., <code>CREATE</code> , <code>ALTER</code>).
DML	Data Manipulation Language – SQL commands for managing data (e.g., <code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code>).
ER	Entity-Relationship – a conceptual model representing entities and their relationships.
RDBMS	Relational Database Management System – software managing relational databases (e.g., MySQL).
CRUD	Create, Read, Update, Delete – basic database operations.
Trigger	A stored SQL procedure automatically executed in response to specific events on a table.
Stored Procedure	A saved collection of SQL statements used to perform tasks in the database.
Function	A user-defined SQL routine that returns a value after processing inputs.
Join	SQL operation combining rows from two or more tables based on related columns.
Nested Query	A query within another SQL query used for filtered or conditional selection.
Aggregate Function	Functions like <code>AVG()</code> , <code>COUNT()</code> , <code>SUM()</code> used to compute summary values.
Primary Key (PK)	A unique identifier for each record in a table.
Foreign Key (FK)	A key linking one table's record to another's primary key to enforce referential integrity.
WildTrack Dashboard	The web-based interface designed for visualization of wildlife data.
Conservation Program	A database entity tracking species protection and awareness efforts.
Habitat	A table entity representing natural environments where species live.
SpeciesID / HabitatID	Unique identifiers used as primary keys in the database schema.
JSON	JavaScript Object Notation – used for data exchange between Flask backend and frontend.
UI/UX	User Interface / User Experience – refers to design and usability of the web dashboard.

Github Link

<https://github.com/KAshyapk07/WildTrack-wildlife-conservation-system>